

## Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

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Query 1: -CREATE DATABASE

```
CREATE DATABASE BankDB1;  
USE BankDB1;
```

Output: -

|   |    |          |                         |                   |           |
|---|----|----------|-------------------------|-------------------|-----------|
| ✓ | 45 | 16:49:53 | CREATE DATABASE BankDB1 | 1 row(s) affected | 0.016 sec |
| ✓ | 46 | 16:49:57 | USE BankDB1             | 0 row(s) affected | 0.000 sec |

Query 2: -CREATE CUSTOMER TABLE

```
CREATE TABLE Customer (  
    Customer_ID INT PRIMARY KEY,  
    Customer_Name VARCHAR(100) NOT NULL,  
    Phone VARCHAR(15),  
    Email VARCHAR(100),  
    City VARCHAR(50)  
);
```

Output: -

|   |    |          |                                             |                   |  |
|---|----|----------|---------------------------------------------|-------------------|--|
| ✓ | 47 | 16:51:31 | CREATE TABLE Customer ( Customer_ID INT ... | 0 row(s) affected |  |
|---|----|----------|---------------------------------------------|-------------------|--|

Query 3: -CREATE ACCOUNT TABLE

```

• CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);

```

Output: -

|   |             |                                            |                   |           |
|---|-------------|--------------------------------------------|-------------------|-----------|
| ✓ | 48 16:56:38 | CREATE TABLE Account ( Account_No INT P... | 0 row(s) affected | 0.063 sec |
|---|-------------|--------------------------------------------|-------------------|-----------|

Query 4: -CREATE BANK TRANSACTION TABLE

```

• CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);

```

Output: -

|   |             |                                              |                   |           |
|---|-------------|----------------------------------------------|-------------------|-----------|
| ✓ | 49 16:59:39 | CREATE TABLE Bank_Transaction ( Transacti... | 0 row(s) affected | 0.079 sec |
|---|-------------|----------------------------------------------|-------------------|-----------|

Query 5: - CREATE LOAN TABLE

```

CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);

```

Output: -

|   |    |          |                                           |                   |           |
|---|----|----------|-------------------------------------------|-------------------|-----------|
| ✓ | 50 | 17:01:41 | CREATE TABLE Loan ( Loan_ID INT PRIMAR... | 0 row(s) affected | 0.109 sec |
|---|----|----------|-------------------------------------------|-------------------|-----------|

Query 6: -INSERT CUSTOMER DATA

```

INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101, 'Aswitha', '9876543218',
'aswitha@gmail.com', 'Hyderabad'),
(102, 'Priyadarshini', '9876543211',
'priya@gmail.com', 'Vijayawada'),
(103, 'Adhritha', '9876543212',
'adhritha@gmail.com', 'Bangalore'),
(104, 'Akshara', '9876543213',
'akshara@gmail.com', 'Chennai'),
(105, 'Sambaiah', '9876543214',
'sam@gmail.com', 'Hyderabad'),
(106, 'Ananya Varshitha', '9876543215',
'ananya@gmail.com', 'Delhi'),
(107, 'Adithya Vardan', '9876543216',
'adithya@gmail.com', 'Mumbai'),
(108, 'Yashwanth', '9876543217',
'yashwanth@gmail.com', 'Pune'),
(109, 'Nihita', '9876543218',
'nihita@gmail.com', 'Hyderabad'),
(110, 'Rushika', '9876543219',
'rushika@gmail.com', 'Vijayawada');

```

Output: -

|   |    |          |                                                |                    |                                 |           |
|---|----|----------|------------------------------------------------|--------------------|---------------------------------|-----------|
| ✓ | 51 | 17:20:23 | INSERT INTO Customer (Customer_ID, Customer... | 10 row(s) affected | Records: 10 Duplicates: 0 Wa... | 0.031 sec |
|---|----|----------|------------------------------------------------|--------------------|---------------------------------|-----------|

#### Query 7: -INSERT ACCOUNT DATA

```
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad'),
(10006, 106, 'Savings', 65000, 'Delhi'),
(10007, 107, 'Current', 150000, 'Mumbai'),
(10008, 108, 'Savings', 35000, 'Pune'),
(10009, 109, 'Savings', 85000, 'Hyderabad'),
(10010, 110, 'Current', 110000, 'Vijayawada');
```

Output: -

```
✓ 52 17:24:47 INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch)
10 row(s) affected Records: 10 Duplicates: 0 Wa... 0.046 sec
```

#### Query 8: -INSERT TRANSACTION DATA

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10001, 'WITHDRAW', 5000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000),
(10006, 'DEPOSIT', 12000),
(10007, 'DEPOSIT', 25000),
(10008, 'WITHDRAW', 5000),
(10009, 'DEPOSIT', 20000),
(10010, 'WITHDRAW', 15000);
```

Output: -

```
✓ 53 17:26:24 INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount)
11 row(s) affected Records: 11 Duplicates: 0 Wa... 0.031 sec
```

Query 9: -INSERT LOAN DATA

```
• INSERT INTO Loan
  (Loan_ID, Customer_ID, Loan_Type,
   Loan_Amount, Interest_Rate)
VALUES
  (501, 101, 'Home Loan', 5000000, 7.5),

  (502, 102, 'Education Loan', 1000000, 6.5),

  (503, 103, 'Car Loan', 800000, 8.2),

  (504, 104, 'Personal Loan', 500000, 10.5),

  (505, 105, 'Home Loan', 4000000, 7.2),

  (506, 106, 'Car Loan', 900000, 8.5),

  (507, 107, 'Business Loan', 3000000, 9.0),

  (508, 109, 'Personal Loan', 600000, 10.0);
```

Output: -

```
✓ 54 17:27:34 INSERT INTO Loan (Loan_ID, Customer_ID, Loa... 8 row(s) affected Records: 8 Duplicates: 0 Wami... 0.047 sec
```

Query 10: - DISPLAY ORIGINAL TABLES

```
SELECT * FROM Customer;
```

|   | Customer_ID | Customer_Name     | Phone      | Email               | City       |
|---|-------------|-------------------|------------|---------------------|------------|
| ▶ | 101         | Aswitha           | 9876543210 | aswitha@gmail.com   | Hyderabad  |
|   | 102         | Priyadarshini     | 9876543211 | priya@gmail.com     | Vijayawada |
|   | 103         | Adhritha          | 9876543212 | adhritha@gmail.com  | Bangalore  |
|   | 104         | Akshara           | 9876543213 | akshara@gmail.com   | Chennai    |
|   | 105         | Sambaiah          | 9876543214 | sam@gmail.com       | Hyderabad  |
|   | 106         | Ananya Varshitha] | 9876543215 | ananya@gmail.com    | Delhi      |
|   | 107         | Adithya Vardan    | 9876543216 | adithya@gmail.com   | Mumbai     |
|   | 108         | Yashwanth         | 9876543217 | yashwanth@gmail.com | Pune       |
|   | 109         | Nihita            | 9876543218 | nihita@gmail.com    | Hyderabad  |
|   | 110         | Rushika           | 9876543219 | rushika@gmail.com   | Vijayawada |
| * | NULL        | NULL              | NULL       | NULL                | NULL       |

```
SELECT * FROM Account;
```

|   | Account_No | Customer_ID | Account_Type | Balance   | Branch     |
|---|------------|-------------|--------------|-----------|------------|
| ▶ | 10001      | 101         | Savings      | 50000.00  | Hyderabad  |
|   | 10002      | 102         | Savings      | 75000.00  | Vijayawada |
|   | 10003      | 103         | Current      | 120000.00 | Bangalore  |
|   | 10004      | 104         | Savings      | 45000.00  | Chennai    |
|   | 10005      | 105         | Current      | 90000.00  | Hyderabad  |
|   | 10006      | 106         | Savings      | 65000.00  | Delhi      |
|   | 10007      | 107         | Current      | 150000.00 | Mumbai     |
|   | 10008      | 108         | Savings      | 35000.00  | Pune       |
|   | 10009      | 109         | Savings      | 85000.00  | Hyderabad  |
|   | 10010      | 110         | Current      | 110000.00 | Vijayawada |
| * | NULL       | NULL        | NULL         | NULL      | NULL       |

```
SELECT * FROM Bank_Transaction;
```

```
SELECT * FROM Loan;
```

|   | Transaction_ID | Account_No | Transaction_Type | Amount   | Transaction_Date    |
|---|----------------|------------|------------------|----------|---------------------|
| ▶ | 1              | 10001      | DEPOSIT          | 10000.00 | 2026-08-15 17:26:24 |
|   | 2              | 10001      | WITHDRAW         | 5000.00  | 2026-08-15 17:26:24 |
|   | 3              | 10002      | DEPOSIT          | 15000.00 | 2026-08-15 17:26:24 |
|   | 4              | 10003      | WITHDRAW         | 20000.00 | 2026-08-15 17:26:24 |
|   | 5              | 10004      | DEPOSIT          | 5000.00  | 2026-08-15 17:26:24 |
|   | 6              | 10005      | WITHDRAW         | 10000.00 | 2026-08-15 17:26:24 |
|   | 7              | 10006      | DEPOSIT          | 12000.00 | 2026-08-15 17:26:24 |
|   | 8              | 10007      | DEPOSIT          | 25000.00 | 2026-08-15 17:26:24 |
|   | 9              | 10008      | WITHDRAW         | 5000.00  | 2026-08-15 17:26:24 |
|   | 10             | 10009      | DEPOSIT          | 20000.00 | 2026-08-15 17:26:24 |
|   | 11             | 10010      | WITHDRAW         | 15000.00 | 2026-08-15 17:26:24 |
| * | NULL           | NULL       | NULL             | NULL     | NULL                |

|   | Loan_ID | Customer_ID | Loan_Type      | Loan_Amount | Interest_Rate |
|---|---------|-------------|----------------|-------------|---------------|
| ▶ | 501     | 101         | Home Loan      | 5000000.00  | 7.50          |
|   | 502     | 102         | Education Loan | 1000000.00  | 6.50          |
|   | 503     | 103         | Car Loan       | 800000.00   | 8.20          |
|   | 504     | 104         | Personal Loan  | 500000.00   | 10.50         |
|   | 505     | 105         | Home Loan      | 4000000.00  | 7.20          |
|   | 506     | 106         | Car Loan       | 900000.00   | 8.50          |
|   | 507     | 107         | Business Loan  | 3000000.00  | 9.00          |
|   | 508     | 109         | Personal Loan  | 600000.00   | 10.00         |
| * | NULL    | NULL        | NULL           | NULL        | NULL          |

Query 11: -CREATE A VIEW FOR ALL CUSTOMERS

```
CREATE VIEW Customer_View AS
SELECT *
FROM Customer;
SELECT * FROM Customer_View;
```

Output: -

|   | Customer_ID | Customer_Name     | Phone      | Email               | City       |
|---|-------------|-------------------|------------|---------------------|------------|
| ▶ | 101         | Aswitha           | 9876543210 | aswitha@gmail.com   | Hyderabad  |
|   | 102         | Priyadarshini     | 9876543211 | priya@gmail.com     | Vijayawada |
|   | 103         | Adhriitha         | 9876543212 | adhriitha@gmail.com | Bangalore  |
|   | 104         | Akshara           | 9876543213 | akshara@gmail.com   | Chennai    |
|   | 105         | Sambaiah          | 9876543214 | sam@gmail.com       | Hyderabad  |
|   | 106         | Ananya Varshitha] | 9876543215 | ananya@gmail.com    | Delhi      |
|   | 107         | Adithya Vardan    | 9876543216 | adithya@gmail.com   | Mumbai     |
|   | 108         | Yashwanth         | 9876543217 | yashwanth@gmail.com | Pune       |
|   | 109         | Nihita            | 9876543218 | nihita@gmail.com    | Hyderabad  |
|   | 110         | Rushika           | 9876543219 | rushika@gmail.com   | Vijayawada |

Customer\_View 5 x

Output

📄 Action Output ▼

|   | #  | Time     | Action                                              |
|---|----|----------|-----------------------------------------------------|
| ✓ | 55 | 17:28:42 | SELECT * FROM Customer LIMIT 0, 1000                |
| ✓ | 56 | 17:29:56 | SELECT * FROM Account LIMIT 0, 1000                 |
| ✓ | 57 | 17:30:37 | SELECT * FROM Bank_Transaction LIMIT 0, 1000        |
| ✓ | 58 | 17:30:56 | SELECT * FROM Loan LIMIT 0, 1000                    |
| ✓ | 59 | 17:33:03 | CREATE VIEW Customer_View AS SELECT * FROM Customer |
| ✓ | 60 | 17:33:07 | SELECT * FROM Customer_View LIMIT 0, 1000           |

Query 12: -HIGH BALANCE ACCOUNT VIEW

```
CREATE VIEW High_Balance_View AS
SELECT
    Account_No,
    Customer_ID,
    Account_Type,
    Balance
FROM Account
WHERE Balance > 100000;
SELECT * FROM High_Balance_View;
```

Output: -

| Result Grid |            |                    |              |           |
|-------------|------------|--------------------|--------------|-----------|
|             |            | Filter Rows:       | Export:      |           |
|             |            | Wrap Cell Content: |              |           |
|             | Account_No | Customer_ID        | Account_Type | Balance   |
| ▶           | 10003      | 103                | Current      | 120000.00 |
|             | 10007      | 107                | Current      | 150000.00 |
|             | 10010      | 110                | Current      | 110000.00 |

  

| High_Balance_View 6 × |          |                                                                                         |  |  |
|-----------------------|----------|-----------------------------------------------------------------------------------------|--|--|
| Output                |          |                                                                                         |  |  |
| Action Output         |          |                                                                                         |  |  |
| #                     | Time     | Action                                                                                  |  |  |
| ✓ 57                  | 17:30:37 | SELECT * FROM Bank_Transaction LIMIT 0, 1000                                            |  |  |
| ✓ 58                  | 17:30:56 | SELECT * FROM Loan LIMIT 0, 1000                                                        |  |  |
| ✓ 59                  | 17:33:03 | CREATE VIEW Customer_View AS SELECT * FROM Customer                                     |  |  |
| ✓ 60                  | 17:33:07 | SELECT * FROM Customer_View LIMIT 0, 1000                                               |  |  |
| ✓ 61                  | 17:35:31 | CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, Account_Type, Balan... |  |  |
| ✓ 62                  | 17:35:34 | SELECT * FROM High_Balance_View LIMIT 0, 1000                                           |  |  |

Query 13: -VIEWS USING JOINS

CUSTOMER + ACCOUNT VIEW

- **CREATE VIEW Customer\_Account\_View AS**  
**SELECT**  
C.Customer\_ID,  
C.Customer\_Name,  
C.City,  
A.Account\_No,  
A.Account\_Type,  
A.Balance,  
A.Branch  
**FROM** Customer C  
**JOIN** Account A  
**ON** C.Customer\_ID = A.Customer\_ID;  
• **SELECT \* FROM Customer\_Account\_View;**

Output: -

|   | Customer_ID | Customer_Name     | City       | Account_No | Account_Type | Balance   | Branch     |
|---|-------------|-------------------|------------|------------|--------------|-----------|------------|
| ▶ | 101         | Aswitha           | Hyderabad  | 10001      | Savings      | 50000.00  | Hyderabad  |
|   | 102         | Priyadarshini     | Vijayawada | 10002      | Savings      | 75000.00  | Vijayawada |
|   | 103         | Adhitha           | Bangalore  | 10003      | Current      | 120000.00 | Bangalore  |
|   | 104         | Akshara           | Chennai    | 10004      | Savings      | 45000.00  | Chennai    |
|   | 105         | Sambaiah          | Hyderabad  | 10005      | Current      | 90000.00  | Hyderabad  |
|   | 106         | Ananya Varshitha] | Delhi      | 10006      | Savings      | 65000.00  | Delhi      |
|   | 107         | Adithya Vardan    | Mumbai     | 10007      | Current      | 150000.00 | Mumbai     |
|   | 108         | Yashwanth         | Pune       | 10008      | Savings      | 35000.00  | Pune       |
|   | 109         | Nihita            | Hyderabad  | 10009      | Savings      | 85000.00  | Hyderabad  |
|   | 110         | Rushika           | Vijayawada | 10010      | Current      | 110000.00 | Vijayawada |

Customer\_Account\_View 7 ×

Output

Action Output

| #    | Time     | Action                                                                                    | Message            |
|------|----------|-------------------------------------------------------------------------------------------|--------------------|
| ✓ 59 | 17:33:03 | CREATE VIEW Customer_View AS SELECT * FROM Customer                                       | 0 row(s) affected  |
| ✓ 60 | 17:33:07 | SELECT * FROM Customer_View LIMIT 0, 1000                                                 | 10 row(s) returned |
| ✓ 61 | 17:35:31 | CREATE VIEW High_Balance_View AS SELECT Account_No, Customer_ID, Account_Type, Balan...   | 0 row(s) affected  |
| ✓ 62 | 17:35:34 | SELECT * FROM High_Balance_View LIMIT 0, 1000                                             | 3 row(s) returned  |
| ✓ 63 | 17:38:10 | CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, C.City, A.... | 0 row(s) affected  |
| ✓ 64 | 17:38:14 | SELECT * FROM Customer_Account_View LIMIT 0, 1000                                         | 10 row(s) returned |

Query 14: -VIEWS USING AGGREGATE FUNCTIONS

AVERAGE ACCOUNT BALANCE

```
CREATE VIEW Average_Account_Balance AS
SELECT
    AVG(Balance) AS Average_Balance
FROM Account;
SELECT * FROM Average_Account_Balance;
```

Output: -

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Result Grid

| Average_Balance |
|-----------------|
| 82500.000000    |

Average\_Account\_Balance 8 x Read Only Context Help

Output

Action Output

| #  | Time     | Action                                        | Message            |
|----|----------|-----------------------------------------------|--------------------|
| 61 | 17:35:31 | CREATE VIEW High_Balance_View AS SELECT...    | 0 row(s) affected  |
| 62 | 17:35:34 | SELECT * FROM High_Balance_View LIMIT 0, 1... | 3 row(s) returned  |
| 63 | 17:38:10 | CREATE VIEW Customer_Account_View AS SEL...   | 0 row(s) affected  |
| 64 | 17:38:14 | SELECT * FROM Customer_Account_View LIMIT...  | 10 row(s) returned |
| 65 | 17:40:36 | CREATE VIEW Average_Account_Balance AS S...   | 0 row(s) affected  |
| 66 | 17:40:39 | SELECT * FROM Average_Account_Balance LI...   | 1 row(s) returned  |

Query 15: -MINIMUM ACCOUNT BALANCE

```
CREATE VIEW Minimum_Balance_View AS
SELECT
    MIN(Balance) AS Minimum_Balance
FROM Account;
SELECT * FROM Minimum_Balance_View;
```

Output: -

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Result Grid

|   | Minimum_Balance |
|---|-----------------|
| ▶ | 35000.00        |

Minimum\_Balance\_View 9 x Read Only Co

Output

Action Output

| #  | Time     | Action                                       | Message            |
|----|----------|----------------------------------------------|--------------------|
| 63 | 17:38:10 | CREATE VIEW Customer_Account_View AS SEL...  | 0 row(s) affected  |
| 64 | 17:38:14 | SELECT * FROM Customer_Account_View LIMIT... | 10 row(s) returned |
| 65 | 17:40:36 | CREATE VIEW Average_Account_Balance AS S...  | 0 row(s) affected  |
| 66 | 17:40:39 | SELECT * FROM Average_Account_Balance LI...  | 1 row(s) returned  |
| 67 | 17:42:06 | CREATE VIEW Minimum_Balance_View AS SEL...   | 0 row(s) affected  |
| 68 | 17:42:09 | SELECT * FROM Minimum_Balance_View LIMIT ... | 1 row(s) returned  |

Query 16: -ACCOUNT TYPE TOTAL BALANCE

```
CREATE VIEW Account_Type_Balance AS
SELECT
    Account_Type,
    SUM(Balance) AS Total_Balance
FROM Account
GROUP BY Account_Type;
SELECT * FROM Account_Type_Balance;
```

Output: -

|   | Account_Type | Total_Balance |
|---|--------------|---------------|
| ▶ | Savings      | 355000.00     |
|   | Current      | 470000.00     |

  

Account\_Type\_Balance 10 x

Output

Action Output

|   | #  | Time     | Action                                       | Message           |
|---|----|----------|----------------------------------------------|-------------------|
| ✓ | 65 | 17:40:36 | CREATE VIEW Average_Account_Balance AS S...  | 0 row(s) affected |
| ✓ | 66 | 17:40:39 | SELECT * FROM Average_Account_Balance LI...  | 1 row(s) returned |
| ✓ | 67 | 17:42:06 | CREATE VIEW Minimum_Balance_View AS SEL...   | 0 row(s) affected |
| ✓ | 68 | 17:42:09 | SELECT * FROM Minimum_Balance_View LIMIT ... | 1 row(s) returned |
| ✓ | 69 | 17:46:06 | CREATE VIEW Account_Type_Balance AS SEL...   | 0 row(s) affected |
| ✓ | 70 | 17:46:09 | SELECT * FROM Account_Type_Balance LIMIT ... | 2 row(s) returned |

Query 17: -HAVING IN VIEWS

BRANCHES WITH TOTAL BALANCE ABOVE 100000

```
CREATE VIEW Rich_Branches AS
SELECT
    Branch,
    SUM(Balance) AS Total_Balance
FROM Account
GROUP BY Branch
HAVING SUM(Balance) > 100000;
SELECT * FROM Rich_Branches;
```

Output: -

|   | Branch     | Total_Balance |
|---|------------|---------------|
| ▶ | Hyderabad  | 225000.00     |
|   | Vijayawada | 185000.00     |
|   | Bangalore  | 120000.00     |
|   | Mumbai     | 150000.00     |

  

Rich\_Branches 11 x

Output

Action Output

|   | #  | Time     | Action                                       | Message               |
|---|----|----------|----------------------------------------------|-----------------------|
| ✓ | 68 | 17:42:09 | SELECT * FROM Minimum_Balance_View LIMIT ... | 1 row(s) returned     |
| ✓ | 69 | 17:46:06 | CREATE VIEW Account_Type_Balance AS SEL...   | 0 row(s) affected     |
| ✓ | 70 | 17:46:09 | SELECT * FROM Account_Type_Balance LIMIT ... | 2 row(s) returned     |
| ✗ | 71 | 17:48:15 | SELECT * FROM Rich_Branches LIMIT 0, 1000    | Error Code: 1146. Tab |
| ✓ | 72 | 17:48:19 | CREATE VIEW Rich_Branches AS SELECT Br...    | 0 row(s) affected     |
| ✓ | 73 | 17:48:21 | SELECT * FROM Rich_Branches LIMIT 0, 1000    | 4 row(s) returned     |

Query 18: -TRANSACTION VIEWS

DEPOSIT TRANSACTION VIEW

```
CREATE VIEW Deposit_Transaction_View AS
SELECT *
FROM Bank_Transaction
WHERE Transaction_Type = 'DEPOSIT';
SELECT * FROM Deposit_Transaction_View;
```

Output: -

| Result Grid        |                |            |                  |          |                     |
|--------------------|----------------|------------|------------------|----------|---------------------|
| Filter Rows:       |                |            |                  |          |                     |
| Export:            |                |            |                  |          |                     |
| Wrap Cell Content: |                |            |                  |          |                     |
|                    | Transaction_ID | Account_No | Transaction_Type | Amount   | Transaction_Date    |
| ▶                  | 1              | 10001      | DEPOSIT          | 10000.00 | 2026-08-15 17:26:24 |
|                    | 3              | 10002      | DEPOSIT          | 15000.00 | 2026-08-15 17:26:24 |
|                    | 5              | 10004      | DEPOSIT          | 5000.00  | 2026-08-15 17:26:24 |
|                    | 7              | 10006      | DEPOSIT          | 12000.00 | 2026-08-15 17:26:24 |
|                    | 8              | 10007      | DEPOSIT          | 25000.00 | 2026-08-15 17:26:24 |
|                    | 10             | 10009      | DEPOSIT          | 20000.00 | 2026-08-15 17:26:24 |

  

Deposit\_Transaction\_View12 x

Output

Action Output

| #    | Time     | Action                                                                                         |
|------|----------|------------------------------------------------------------------------------------------------|
| ✓ 70 | 17:46:09 | SELECT * FROM Account_Type_Balance LIMIT 0, 1000                                               |
| ✗ 71 | 17:48:15 | SELECT * FROM Rich_Branches LIMIT 0, 1000                                                      |
| ✓ 72 | 17:48:19 | CREATE VIEW Rich_Branches AS SELECT Branch, SUM(Balance) AS Total_Balance FROM Account ...     |
| ✓ 73 | 17:48:21 | SELECT * FROM Rich_Branches LIMIT 0, 1000                                                      |
| ✓ 74 | 17:50:13 | CREATE VIEW Deposit_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction_Ty... |
| ✓ 75 | 17:50:16 | SELECT * FROM Deposit_Transaction_View LIMIT 0, 1000                                           |

Query 19: -VIEWS WITH ORDER BY

CUSTOMERS ORDERED BY NAME

```
CREATE VIEW Customer_Name_View AS
SELECT
    Customer_ID,
    Customer_Name,
    City
FROM Customer
ORDER BY Customer_Name;
SELECT * FROM Customer_Name_View;
```

Output: -

|   | Customer_ID | Customer_Name     | City       |
|---|-------------|-------------------|------------|
| ▶ | 103         | Adhritha          | Bangalore  |
|   | 107         | Adithya Vardan    | Mumbai     |
|   | 104         | Akshara           | Chennai    |
|   | 106         | Ananya Varshitha] | Delhi      |
|   | 101         | Aswitha           | Hyderabad  |
|   | 109         | Nihita            | Hyderabad  |
|   | 102         | Priyadarshini     | Vijayawada |
|   | 110         | Rushika           | Vijayawada |
|   | 105         | Sambaiah          | Hyderabad  |
|   | 108         | Yashwanth         | Pune       |

  

Customer\_Name\_View 13 × ⓘ Read

Output

📄

Action Output

▼

|   | #  | Time     | Action                                        | Message            |
|---|----|----------|-----------------------------------------------|--------------------|
| ✓ | 72 | 17:48:19 | CREATE VIEW Rich_Branches AS SELECT Br...     | 0 row(s) affected  |
| ✓ | 73 | 17:48:21 | SELECT * FROM Rich_Branches LIMIT 0, 1000     | 4 row(s) returned  |
| ✓ | 74 | 17:50:13 | CREATE VIEW Deposit_Transaction_View AS S...  | 0 row(s) affected  |
| ✓ | 75 | 17:50:16 | SELECT * FROM Deposit_Transaction_View LIM... | 6 row(s) returned  |
| ✓ | 76 | 17:52:27 | CREATE VIEW Customer_Name_View AS SELE...     | 0 row(s) affected  |
| ✓ | 77 | 17:52:30 | SELECT * FROM Customer_Name_View LIMIT 0,...  | 10 row(s) returned |

Query 20: -VIEW WITH CALCULATED COLUMN

LOAN TOTAL AMOUNT VIEW

```

CREATE VIEW Loan_Total_Amount_View AS
SELECT
    Loan_ID,
    Customer_ID,
    Loan_Type,
    Loan_Amount,
    Interest_Rate,
    Loan_Amount +
    (Loan_Amount * Interest_Rate / 100)
    AS Total_Amount
FROM Loan;
SELECT * FROM Loan_Total_Amount_View;

```

Output: -

|   | Loan_ID | Customer_ID | Loan_Type      | Loan_Amount | Interest_Rate | Total_Amount      |
|---|---------|-------------|----------------|-------------|---------------|-------------------|
| ▶ | 501     | 101         | Home Loan      | 5000000.00  | 7.50          | 5375000.000000000 |
|   | 502     | 102         | Education Loan | 1000000.00  | 6.50          | 1065000.000000000 |
|   | 503     | 103         | Car Loan       | 800000.00   | 8.20          | 865600.000000000  |
|   | 504     | 104         | Personal Loan  | 500000.00   | 10.50         | 552500.000000000  |
|   | 505     | 105         | Home Loan      | 4000000.00  | 7.20          | 4288000.000000000 |
|   | 506     | 106         | Car Loan       | 900000.00   | 8.50          | 976500.000000000  |
|   | 507     | 107         | Business Loan  | 3000000.00  | 9.00          | 3270000.000000000 |
|   | 508     | 109         | Personal Loan  | 600000.00   | 10.00         | 660000.000000000  |

Loan\_Total\_Amount\_View 14 x

Output

Action Output

| #    | Time     | Action                                                                                   |
|------|----------|------------------------------------------------------------------------------------------|
| ✓ 74 | 17:50:13 | CREATE VIEW Deposit_Transaction_View AS SELECT * FROM Bank_Transaction WHERE Transaction |
| ✓ 75 | 17:50:16 | SELECT * FROM Deposit_Transaction_View LIMIT 0, 1000                                     |
| ✓ 76 | 17:52:27 | CREATE VIEW Customer_Name_View AS SELECT Customer_ID, Customer_Name, City FROM C         |
| ✓ 77 | 17:52:30 | SELECT * FROM Customer_Name_View LIMIT 0, 1000                                           |
| ✓ 78 | 17:55:12 | CREATE VIEW Loan_Total_Amount_View AS SELECT Loan_ID, Customer_ID, Loan_Type, L          |
| ✓ 79 | 17:55:15 | SELECT * FROM Loan_Total_Amount_View LIMIT 0, 1000                                       |

Query 21: -USING VIEW WITH QUERIES

DISPLAY CUSTOMER ACCOUNTS ORDERED BY BALANCE

```

SELECT *
FROM Customer_Account_View
ORDER BY Balance DESC;

```

Output: -

|   | Customer_ID | Customer_Name     | City       | Account_No | Account_Type | Balance   | Branch     |
|---|-------------|-------------------|------------|------------|--------------|-----------|------------|
| ▶ | 107         | Adithya Vardan    | Mumbai     | 10007      | Current      | 150000.00 | Mumbai     |
|   | 103         | Adhritha          | Bangalore  | 10003      | Current      | 120000.00 | Bangalore  |
|   | 110         | Rushika           | Vijayawada | 10010      | Current      | 110000.00 | Vijayawada |
|   | 105         | Sambaiah          | Hyderabad  | 10005      | Current      | 90000.00  | Hyderabad  |
|   | 109         | Nihita            | Hyderabad  | 10009      | Savings      | 85000.00  | Hyderabad  |
|   | 102         | Priyadarshini     | Vijayawada | 10002      | Savings      | 75000.00  | Vijayawada |
|   | 106         | Ananya Varshitha] | Delhi      | 10006      | Savings      | 65000.00  | Delhi      |
|   | 101         | Aswitha           | Hyderabad  | 10001      | Savings      | 50000.00  | Hyderabad  |
|   | 104         | Akshara           | Chennai    | 10004      | Savings      | 45000.00  | Chennai    |
|   | 108         | Yashwanth         | Pune       | 10008      | Savings      | 35000.00  | Pune       |

Query 22: -INSERT THROUGH VIEW

```
CREATE VIEW Simple_Account_View AS
SELECT
    Account_No,
    Customer_ID,
    Account_Type,
    Balance,
    Branch
FROM Account;
INSERT INTO Simple_Account_View
VALUES
(10011, 101, 'Savings', 55000, 'Hyderabad');
SELECT *
FROM Account;
```

Output: -

|   | Account_No | Customer_ID | Account_Type | Balance   | Branch     |
|---|------------|-------------|--------------|-----------|------------|
| ▶ | 10001      | 101         | Savings      | 50000.00  | Hyderabad  |
|   | 10002      | 102         | Savings      | 75000.00  | Vijayawada |
|   | 10003      | 103         | Current      | 120000.00 | Bangalore  |
|   | 10004      | 104         | Savings      | 45000.00  | Chennai    |
|   | 10005      | 105         | Current      | 90000.00  | Hyderabad  |
|   | 10006      | 106         | Savings      | 65000.00  | Delhi      |
|   | 10007      | 107         | Current      | 150000.00 | Mumbai     |
|   | 10008      | 108         | Savings      | 35000.00  | Pune       |
|   | 10009      | 109         | Savings      | 85000.00  | Hyderabad  |
|   | 10010      | 110         | Current      | 110000.00 | Vijayawada |
|   | 10011      | 101         | Savings      | 55000.00  | Hyderabad  |
| * | NULL       | NULL        | NULL         | NULL      | NULL       |

|   |    |          |                                              |                    |
|---|----|----------|----------------------------------------------|--------------------|
| ✓ | 85 | 17:59:56 | SELECT * FROM Account WHERE Account_No ...   | 1 row(s) returned  |
| ✓ | 86 | 18:02:14 | CREATE VIEW Simple_Account_View AS SELE...   | 0 row(s) affected  |
| ✓ | 87 | 18:02:25 | INSERT INTO Simple_Account_View VALUES (1... | 1 row(s) affected  |
| ✓ | 88 | 18:02:29 | SELECT * FROM Account LIMIT 0, 1000          | 11 row(s) returned |

Query 23: -DELETE THROUGH VIEW

```
DELETE FROM Simple_Account_View
WHERE Account_No = 10011;
SELECT *
FROM Account;
```

Output: -

|   | Account_No | Customer_ID | Account_Type | Balance   | Branch     |
|---|------------|-------------|--------------|-----------|------------|
| ▶ | 10001      | 101         | Savings      | 50000.00  | Hyderabad  |
|   | 10002      | 102         | Savings      | 75000.00  | Vijayawada |
|   | 10003      | 103         | Current      | 120000.00 | Bangalore  |
|   | 10004      | 104         | Savings      | 45000.00  | Chennai    |
|   | 10005      | 105         | Current      | 90000.00  | Hyderabad  |
|   | 10006      | 106         | Savings      | 65000.00  | Delhi      |
|   | 10007      | 107         | Current      | 150000.00 | Mumbai     |
|   | 10008      | 108         | Savings      | 35000.00  | Pune       |
|   | 10009      | 109         | Savings      | 85000.00  | Hyderabad  |
|   | 10010      | 110         | Current      | 110000.00 | Vijayawada |
| • | NULL       | NULL        | NULL         | NULL      | NULL       |

Result Grid

Form Editor

Field Types

Account 19 x

Apply

Output

Action Output ▼

|   | #  | Time     | Action                                       | Message            |
|---|----|----------|----------------------------------------------|--------------------|
| ✓ | 85 | 17:59:56 | SELECT * FROM Account WHERE Account_No ...   | 1 row(s) returned  |
| ✓ | 86 | 18:02:14 | CREATE VIEW Simple_Account_View AS SELE...   | 0 row(s) affected  |
| ✓ | 87 | 18:02:25 | INSERT INTO Simple_Account_View VALUES (1... | 1 row(s) affected  |
| ✓ | 88 | 18:02:29 | SELECT * FROM Account LIMIT 0, 1000          | 11 row(s) returned |
| ✓ | 89 | 18:03:55 | DELETE FROM Simple_Account_View WHERE ...    | 1 row(s) affected  |
| ✓ | 90 | 18:04:01 | SELECT * FROM Account LIMIT 0, 1000          | 10 row(s) returned |

Query 24: -VIEW INFORMATION

SHOW FULL TABLES

WHERE TABLE\_TYPE = 'VIEW';

Output: -

|   | Tables_in_bankdb1        | Table_type |
|---|--------------------------|------------|
| ▶ | account_type_balance     | VIEW       |
|   | average_account_balance  | VIEW       |
|   | customer_account_view    | VIEW       |
|   | customer_name_view       | VIEW       |
|   | customer_view            | VIEW       |
|   | deposit_transaction_view | VIEW       |
|   | high_balance_view        | VIEW       |
|   | loan_total_amount_view   | VIEW       |
|   | minimum_balance_view     | VIEW       |
|   | rich_branches            | VIEW       |
|   | simple_account_view      | VIEW       |